

Requirements Generation Summary

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Product Category: Robotics – Humanoid robot

61

Total Requirements

1

Systems

3

Subsystems

System Description

A high-dexterity, bio-mimetic end-effector designed for the Optimus 3 humanoid platform. The system provides 22 degrees of freedom (DoF) with integrated tactile sensing, high-torque density actuation, and real-time closed-loop control. It interfaces with the Optimus 3 forearm via a standardized mechanical coupling and a high-bandwidth EtherCAT communication bus. Operational context includes complex manipulation tasks in unstructured environments, requiring human-equivalent force application and precision. Non-functional requirements prioritize ISO 10218-1 safety compliance, a 5,000-hour MTBF, and thermal management within a 45°C surface temperature limit for human interaction.

Optimus 3 Robotic Hand System

System-Level Requirements

OPT3-HND-001

Functional

The Optimus 3 Robotic Hand System SHALL provide 22 independent degrees of freedom (DoF) distributed across five digits and a palm to enable bio-mimetic manipulation of human-scale objects.

Rationale: To achieve the dexterity required for unstructured environments and human-equivalent tasks, the hand must match the kinematic complexity of a human hand.

Verification: Inspection

Acceptance: Physical count of independent actuators and kinematic joints confirms 22 distinct degrees of freedom.

OPT3-HND-002

Performance

The system SHALL integrate tactile sensors on all fingertip and palm contact surfaces capable of measuring normal forces with a resolution of 0.1 N and a minimum sampling frequency of 500 Hz.

Rationale: High-resolution tactile feedback is necessary for closed-loop control during delicate manipulation and to prevent slippage or damage to objects.

Verification: Test

Acceptance: Sensor data logs demonstrate 0.1 N sensitivity and 500 Hz update rates across all specified surfaces during calibrated force application.

OPT3-HND-003

Safety

The system SHALL maintain all external surface temperatures at or below 45°C during continuous operation at 50% rated duty cycle in a 25°C ambient environment.

Rationale: Compliance with safety standards for human-robot interaction requires surface temperatures to remain below the threshold for skin discomfort or burns.

Verification: Test

Acceptance: Thermal imaging and contact probes confirm no external point exceeds 45°C after 2 hours of continuous operation at the specified duty cycle.

The system SHALL communicate with the Optimus 3 forearm controller via an EtherCAT interface supporting a Cyclic Synchronous Position (CSP) update rate of 1 kHz with a maximum jitter of 10 microseconds.

Rationale: High-bandwidth, low-latency communication is critical for real-time stability and synchronization of 22 DoF in complex manipulation tasks.

Verification: Test

Acceptance: Network analyzer logs confirm 1 kHz packet frequency and jitter within the 10-microsecond tolerance over a 24-hour test period.

The system SHALL achieve a Mean Time Between Failures (MTBF) of no less than 5,000 hours when operated within the nominal load profile and MIL-STD-810H environmental conditions.

Rationale: High reliability is required to minimize downtime and maintenance costs for the humanoid platform in industrial or service contexts.

Verification: Analysis

Acceptance: Reliability Block Diagram (RBD) and Accelerated Life Testing (ALT) data statistically project an MTBF $\geq$ 5,000 hours with a 90% confidence interval.

The system SHALL implement force-limiting safety functions that trigger a Category 0 stop in accordance with ISO 10218-1 if detected contact forces exceed 150 N.

Rationale: Ensures the system operates safely around humans by preventing excessive force application during collisions or malfunctions.

Verification: Test

Acceptance: The system executes a power-off stop within 50ms when an external force of 151 N is applied to any digit.

Actuation and Kinematics Subsystem

ACT-001

Functional

The Actuation and Kinematics Subsystem SHALL provide 22 independent degrees of freedom (DoF) distributed across four fingers and one opposable thumb to enable bio-mimetic manipulation.

Rationale: To meet the system-level requirement for high-dexterity and human-equivalent manipulation capabilities.

Verification: Inspection

Acceptance: Physical count and kinematic model verification confirm 22 independently controllable joints.

ACT-002

Performance

The Actuation and Kinematics Subsystem SHALL exert a continuous fingertip pinch force of at least 35 Newtons (N) for each finger and 50 Newtons (N) for the thumb.

Rationale: Ensures the hand can perform human-equivalent force application for heavy-duty manipulation tasks.

Verification: Test

Acceptance: Load cell measurements at the fingertips show sustained force $\geq 35\text{N}$ (fingers) and $\geq 50\text{N}$ (thumb) for 60 seconds without thermal shutdown.

ACT-003

Environmental

The Actuation and Kinematics Subsystem SHALL maintain an external surface temperature not exceeding 45°C during continuous operation at 50% of rated torque in a 25°C ambient environment.

Rationale: Required for safety compliance during human-robot interaction and to prevent damage to integrated tactile sensors.

Verification: Test

Acceptance: Thermal imaging confirms no external point on the hand exceeds 45°C after 2 hours of continuous cyclic operation.

ACT-004

Performance

The hybrid transmission system SHALL exhibit a maximum mechanical backlash of less than 0.5 degrees at the joint output.

Rationale: Low backlash is critical for high-precision closed-loop control and stability of the EtherCAT communication loop.

Verification: Test

Acceptance: Angular displacement measurements at the joint output show < 0.5 degrees of play when the input is locked.

ACT-005

Reliability

The Actuation and Kinematics Subsystem SHALL achieve a Mean Time Between Failures (MTBF) of at least 5,000 hours under nominal operating conditions.

Rationale: Directly traces to the system-level reliability requirement for the Optimus 3 platform.

Verification: Analysis

Acceptance: Reliability block diagram (RBD) and accelerated life testing data validate a 5,000-hour MTBF at a 90% confidence level.

ACT-006

Interface

The Actuation and Kinematics Subsystem SHALL provide joint position and torque feedback to the Control Subsystem at a minimum frequency of 1,000 Hz via the internal bus.

Rationale: High-frequency feedback is necessary for real-time closed-loop control and tactile-responsive manipulation.

Verification: Test

Acceptance: Oscilloscope or bus analyzer confirms data packets for all 22 DoF are delivered to the control interface every 1ms.

High-Torque Micro-BLDC Motors

REQ-BLDC-001

Performance

The Micro-BLDC motor assembly SHALL provide a minimum continuous output torque of 120 mNm and a peak torque of 350 mNm at the gearhead output shaft while maintaining a total assembly mass of less than 30 grams.

Rationale: High torque-to-weight ratio is critical for achieving human-equivalent grip force within the slim form factor of a finger assembly as specified in the Actuation Subsystem requirements.

Verification: Test

Acceptance: Dynamometer testing confirms continuous torque ≥ 120 mNm and peak torque ≥ 350 mNm; mass measurement confirms < 30 g.

REQ-BLDC-002

Functional

The Micro-BLDC motor SHALL incorporate an integrated 10k Ω NTC thermistor embedded within the stator windings to provide real-time temperature telemetry with an accuracy of $\pm 2^{\circ}\text{C}$.

Rationale: Thermal monitoring is required to prevent winding failure and to ensure the hand surface temperature does not exceed the 45 $^{\circ}\text{C}$ safety limit for human interaction.

Verification: Test

Acceptance: Telemetry output matches calibrated external thermal probe readings within $\pm 2^{\circ}\text{C}$ across a range of 25 $^{\circ}\text{C}$ to 100 $^{\circ}\text{C}$.

REQ-BLDC-003

Performance

The integrated planetary gearhead SHALL exhibit a mechanical backlash of no more than 1.0 degree at the output shaft under no-load conditions.

Rationale: Low backlash is essential for the high-dexterity and precision manipulation requirements of the Optimus 3 system, particularly for fine motor tasks.

Verification: Inspection

Acceptance: Angular measurement of output shaft play while the motor rotor is locked shows ≤ 1.0 degree of movement.

REQ-BLDC-004

Reliability

The Micro-BLDC motor and gearhead assembly SHALL achieve a Mean Time Between Failures (MTBF) of at least 5,000 hours under a nominal operating profile of 20% duty cycle.

Rationale: Directly traces to the system-level reliability requirement for the Optimus 3 Robotic Hand System.

Verification: Analysis

Acceptance: Reliability block diagram (RBD) and accelerated life testing data demonstrate a 95% confidence level of meeting the 5,000-hour MTBF.

REQ-BLDC-005

Environmental

The motor assembly SHALL maintain structural integrity and operational performance after exposure to vibration profiles defined in MIL-STD-810H, Method 514.8.

Rationale: Ensures the component can withstand the operational shocks and vibrations typical of mobile humanoid platforms in unstructured environments.

Verification: Test

Acceptance: Post-vibration test inspection shows no mechanical deformation and motor performance remains within $\pm 5\%$ of pre-test torque/speed specifications.

REQ-BLDC-006

Regulatory

The motor assembly SHALL be constructed using materials that are 100% RoHS and REACH compliant.

Rationale: Mandatory regulatory compliance for global distribution and environmental safety as specified in the system assumptions.

Verification: Inspection

Acceptance: Provision of valid Certificates of Conformance (CoC) and Material Safety Data Sheets (MSDS) for all sub-components.

Synthetic Fiber Tendon Assembly

REQ-SFTA-001

Performance

The Synthetic Fiber Tendon SHALL withstand a peak tensile load of 500 N without exceeding the material's elastic limit.

Rationale: Ensures the tendon can transmit the maximum torque generated by the palm-mounted motors during high-force grasping tasks without permanent deformation.

Verification: Test

Acceptance: Tendon maintains original length (within 0.05% tolerance) after 10 cycles of 500 N loading.

REQ-SFTA-002

Performance

The tendon assembly SHALL exhibit a creep rate of less than 0.2% total elongation over 1,000 hours when subjected to a constant tension of 150 N at 25°C.

Rationale: HMPE is susceptible to creep; limiting this is critical for maintaining joint calibration and control precision over the system's lifespan.

Verification: Test

Acceptance: Measured elongation is < 0.2% after 1,000 hours of continuous loading in a controlled environment.

REQ-SFTA-003

Performance

The dynamic coefficient of friction between the HMPE tendon and the internal conduit surface SHALL NOT exceed 0.08.

Rationale: Low friction is required to maximize torque transmission efficiency and minimize heat generation within the finger structure.

Verification: Test

Acceptance: Friction testing using a capstan method confirms $\mu \leq 0.08$ across the full range of motion.

REQ-SFTA-004

Reliability

The tendon assembly SHALL maintain functional integrity for a minimum of 5,000,000 flexion-extension cycles at a nominal load of 100 N.

Rationale: Directly supports the system-level requirement for a 5,000-hour MTBF in high-duty cycle applications.

Verification: Test

Acceptance: No visible fraying or loss of tensile strength > 10% after 5 million cycles.

REQ-SFTA-005

Interface

The tendon SHALL be capable of operating through a minimum bend radius of 1.5 mm without structural degradation.

Rationale: The slim form factor of the humanoid phalanges requires tight routing through the finger joints.

Verification: Inspection

Acceptance: Verification that the tendon maintains its rated break strength after being routed through a 1.5 mm radius guide.

REQ-SFTA-006

Regulatory

The tendon and conduit materials SHALL be compliant with RoHS 3 (Directive 2015/863) and REACH regulations.

Rationale: Ensures environmental compliance and safety for human-robot interaction as specified in the system context.

Verification: Analysis

Acceptance: Supplier Material Data Sheets (MSDS) and Certificates of Compliance (CoC) confirm absence of restricted substances.

Sensory Perception Subsystem

SPS-REQ-001

Performance

The Sensory Perception Subsystem SHALL provide tactile sensing across the palmar and fingertip surfaces with a spatial resolution of no less than 4.0 mm² per taxel and a pressure sensitivity range of 5 kPa to 800 kPa.

Rationale: High-resolution tactile feedback is required for the Optimus 3 to perform complex manipulation of small objects and to detect slip in unstructured environments.

Verification: Test

Acceptance: Verification via calibrated pressure application showing detection at 5 kPa and distinct taxel activation at 2mm spacing.

SPS-REQ-002

Functional

The Sensory Perception Subsystem SHALL measure the absolute angular position of all 22 joints using encoders with a minimum resolution of 16-bit and a repeatability of ± 0.05 degrees.

Rationale: Precise joint position feedback is critical for the 22 DoF closed-loop control system to achieve human-equivalent dexterity and kinematic accuracy.

Verification: Test

Acceptance: Encoder output must maintain ± 0.05 degree repeatability over 1,000 cycles of full-range motion.

SPS-REQ-003

Performance

The Sensory Perception Subsystem SHALL measure 6-axis force and torque at the wrist interface with a minimum force range of ± 200 N (F_x , F_y , F_z) and torque range of ± 10 Nm (T_x , T_y , T_z) at a sampling frequency of 1000 Hz.

Rationale: The wrist F/T sensor is the primary feedback mechanism for interaction force control and safety during heavy-duty manipulation tasks.

Verification: Test

Acceptance: Sensor must report values within 1% of applied load across the full specified range at a 1ms update rate.

SPS-REQ-004

Performance

The Sensory Perception Subsystem SHALL aggregate and transmit pre-processed sensor packets to the EtherCAT bus with a total processing latency not exceeding 250 microseconds.

Rationale: To maintain the stability of the high-bandwidth control loop, the delay between physical sensing and data availability on the bus must be minimized.

Verification: Test

Acceptance: Oscilloscope measurement of the time delta between a sensor stimulus and the corresponding EtherCAT frame transmission.

SPS-REQ-005

Safety

The Sensory Perception Subsystem SHALL include thermal sensors at each actuator cluster to provide real-time temperature data with an accuracy of $\pm 1.0^{\circ}\text{C}$ to the control system.

Rationale: Thermal monitoring is required to ensure the system does not exceed the 45°C surface temperature limit specified in the parent requirements for human interaction safety.

Verification: Test

Acceptance: Thermal readings must be within 1.0°C of a calibrated reference probe during a ramp-up test from 20°C to 60°C .

SPS-REQ-006

Reliability

The Sensory Perception Subsystem SHALL maintain full operational calibration and functionality for a minimum Mean Time Between Failures (MTBF) of 5,000 hours under nominal operating conditions.

Rationale: Directly traces to the parent system reliability requirement to ensure long-term deployment without frequent maintenance.

Verification: Analysis

Acceptance: Reliability block diagram (RBD) and FMEA analysis demonstrating a calculated MTBF $\geq 5,000$ hours.

Multi-Modal Tactile Skin

REQ-TS-001

Performance

The Multi-Modal Tactile Skin SHALL provide a spatial resolution of no less than 2.0 mm between sensing taxels in the fingertip regions and 5.0 mm in the palm region.

Rationale: High-resolution pressure mapping is required for the identification of small objects and texture recognition during manipulation tasks.

Verification: Inspection

Acceptance: Physical measurement of taxel pitch confirms ≤ 2.0 mm at fingertips and ≤ 5.0 mm at the palm.

REQ-TS-002

Performance

The tactile skin SHALL detect normal forces within a range of 0.1 N to 50.0 N per taxel with a minimum sensitivity of 0.05 N.

Rationale: The sensor must be sensitive enough for delicate touch while robust enough to handle the high-torque density actuation of the Optimus 3 hand.

Verification: Test

Acceptance: Calibration tests demonstrate linear or compensable response from 0.1 N to 50.0 N with 0.05 N detectable increments.

REQ-TS-003

Functional

The tactile skin SHALL measure shear forces in the X and Y axes with a minimum range of ± 10 N and a resolution of 0.1 N.

Rationale: Shear force detection is critical for identifying incipient slip and adjusting grip force in real-time closed-loop control.

Verification: Test

Acceptance: Application of tangential loads results in X/Y vector data with $< 10\%$ cross-talk between axes.

REQ-TS-004

Performance

The tactile sensor array SHALL output data to the Sensory Perception Subsystem at a sampling frequency of at least 1 kHz.

Rationale: High-frequency feedback is necessary to meet the low-latency requirements of the EtherCAT communication bus and real-time control loop.

Verification: Test

Acceptance: Oscilloscope or bus analyzer confirms data packets are delivered at intervals of ≤ 1 ms.

REQ-TS-005

Safety

The silicone-based artificial skin SHALL maintain an external surface temperature below 45°C during continuous operation in a 25°C ambient environment.

Rationale: Ensures compliance with system-level safety requirements for human-robot interaction and prevents degradation of the piezoresistive elements.

Verification: Test

Acceptance: Thermal imaging confirms surface temperature does not exceed 45°C after 4 hours of continuous operation.

REQ-TS-006

Reliability

The tactile skin SHALL withstand 1,000,000 load cycles at 25 N normal force without a shift in baseline resistance exceeding 15%.

Rationale: Supports the system-level reliability requirement of a 5,000-hour MTBF in unstructured environments.

Verification: Test

Acceptance: Post-cycling calibration shows baseline resistance drift is $\leq 15\%$ of the initial value.

REQ-TS-007

Regulatory

The tactile skin assembly SHALL be compliant with RoHS and REACH standards for hazardous substances.

Rationale: Mandatory regulatory compliance as specified in the system-level assumptions.

Verification: Analysis

Acceptance: Manufacturer Material Declaration (MMD) or third-party lab report confirms compliance.

Absolute Magnetic Encoders

OPT3-ENC-001

Performance

The absolute magnetic encoder SHALL provide a minimum angular resolution of 14 bits (16,384 counts per revolution) over a full 360-degree rotation.

Rationale: High-resolution feedback is critical for the 22 DoF dexterity required for human-equivalent manipulation and precise closed-loop control.

Verification: Test

Acceptance: Verification that the encoder output increments by exactly 16,384 unique values over one mechanical revolution.

OPT3-ENC-002

Functional

The encoder SHALL output valid absolute position data via the BiSS-C protocol within 50 milliseconds of system power-on without requiring joint movement.

Rationale: Eliminating homing sequences is essential for safety and immediate operational readiness in unstructured environments where movement upon startup could cause collisions.

Verification: Demonstration

Acceptance: Data stream validation showing stable, accurate position bits on the EtherCAT bus within 50ms of VCC application.

OPT3-ENC-003

Interface

The encoder interface SHALL support a BiSS-C clock frequency of up to 10 MHz to ensure compatibility with the high-bandwidth requirements of the Sensory Perception Subsystem.

Rationale: High clock speeds are necessary to minimize latency in the 22-joint control loop and maintain the real-time performance of the Optimus 3 platform.

Verification: Test

Acceptance: Successful data transmission and parity check at a 10 MHz clock rate using the target microcontroller/FPGA.

OPT3-ENC-004

Performance

The encoder assembly SHALL maintain an absolute accuracy error of less than ± 0.1 degrees across the full operating temperature range of -10°C to $+60^{\circ}\text{C}$.

Rationale: Thermal drift must be minimized to ensure the hand's kinematic model remains accurate during prolonged operation and high-torque tasks.

Verification: Test

Acceptance: Measured angular error remains within ± 0.1 degrees when compared to a calibrated reference standard across the specified temperature ramp.

OPT3-ENC-005

Environmental

The encoder SHALL withstand mechanical shocks of up to 40g (11ms sawtooth) in accordance with MIL-STD-810H, Method 516.8, without loss of signal or mechanical displacement.

Rationale: Humanoid hands are subject to frequent impacts during grasping and environmental interaction; the feedback system must be robust against these shocks.

Verification: Test

Acceptance: Post-shock inspection and functional test showing no degradation in signal quality or physical alignment.

OPT3-ENC-006

Regulatory

The encoder electronics SHALL be RoHS and REACH compliant and utilize low-outgassing materials suitable for integrated robotic assemblies.

Rationale: Ensures regulatory compliance and prevents contamination of sensitive tactile sensors within the enclosed hand structure.

Verification: Inspection

Acceptance: Review of manufacturer Material Safety Data Sheets (MSDS) and Certificates of Conformance (CoC).

Embedded Control and Communication Subsystem

ECC-REQ-001

Performance

The Embedded Control Subsystem SHALL execute the low-level PID control loops for all 22 degrees of freedom (DoF) simultaneously at a minimum update frequency of 1 kHz.

Rationale: High-frequency control is necessary to maintain stability and responsiveness for high-dexterity manipulation in unstructured environments.

Verification: Test

Acceptance: Oscilloscope or software profiling confirms all 22 PID loops complete within a 1ms window with less than 5% CPU jitter.

ECC-REQ-002

Interface

The SoC SHALL communicate with the Optimus 3 forearm controller via EtherCAT with a synchronized cycle time of 500 microseconds and a maximum jitter of 1 microsecond.

Rationale: Real-time synchronization is required for coordinated whole-body control and to meet the high-bandwidth requirements of the Optimus 3 platform.

Verification: Test

Acceptance: EtherCAT network analyzer confirms a stable 500 μ s cycle time and jitter within +/- 1 μ s over a 24-hour continuous operation period.

ECC-REQ-003

Safety

The subsystem SHALL implement a safety-rated torque limiting function that cuts power to the actuators within 10 milliseconds if the measured torque exceeds the ISO 10218-1 defined safety thresholds.

Rationale: Ensures compliance with ISO 10218-1 for human-robot collaboration and prevents mechanical damage during collisions.

Verification: Test

Acceptance: Simulated over-torque events result in motor power cutoff in <10ms across all test cases.

ECC-REQ-004

Performance

The SoC SHALL fuse data from the 22 motor encoders and the integrated tactile sensor array with a total processing latency of less than 2 milliseconds.

Rationale: Low-latency sensor fusion is critical for the closed-loop tactile feedback required for human-equivalent precision.

Verification: Analysis

Acceptance: Data processing pipeline analysis and hardware-in-the-loop (HIL) testing show end-to-end latency from sensor interrupt to fused data availability is <2ms.

ECC-REQ-005

Environmental

The Embedded Control Subsystem SHALL monitor internal SoC and driver temperatures and initiate thermal throttling if the internal temperature exceeds 80°C to maintain an external surface temperature below 45°C.

Rationale: Directly traces to the parent requirement for human interaction safety and thermal management.

Verification: Test

Acceptance: Thermal imaging confirms the hand's external surface does not exceed 45°C while the SoC is under 100% load in a 25°C ambient environment.

ECC-REQ-006

Reliability

The subsystem electronics SHALL be designed with a calculated Mean Time Between Failures (MTBF) of no less than 15,000 hours.

Rationale: To meet the system-level MTBF of 5,000 hours, individual electronic subsystems must exhibit higher reliability to account for mechanical wear and other failure points.

Verification: Analysis

Acceptance: Reliability Workbench or similar MIL-HDBK-217F analysis demonstrates a calculated MTBF $\geq 15,000$ hours.

Real-Time Processing Unit (RPU)

RPU-REQ-001

Performance

The RPU SHALL execute the inverse kinematics (IK) solver for all 22 degrees of freedom (DoF) within a control loop cycle time of no more than 500 microseconds.

Rationale: To maintain stable real-time control and smooth motion of the 22-DoF hand, the IK calculations must be completed well within the 1kHz (1ms) system update rate.

Verification: Test

Acceptance: Profiling of the IK task on the Cortex-M7 shows a worst-case execution time (WCET) $\leq 500\mu\text{s}$ over 1,000,000 iterations.

RPU-REQ-002**Safety**

The RPU SHALL implement a safety-rated torque monitoring function that triggers a Category 0 stop (per IEC 60204-1) within 10ms if any joint torque exceeds the dynamic limit defined in the safety configuration.

Rationale: Ensures compliance with ISO 10218-1 for human-robot interaction and prevents mechanical damage during collisions.

Verification: Test

Acceptance: Measured latency from torque threshold violation to motor power-off signal is $\leq 10\text{ms}$.

RPU-REQ-003**Functional**

The RPU SHALL perform cubic spline trajectory interpolation to generate joint position setpoints at a frequency of 1 kHz.

Rationale: High-frequency interpolation is required to prevent jitter and ensure bio-mimetic smoothness in finger movement.

Verification: Test

Acceptance: Oscilloscope or logic analyzer verification of 1kHz setpoint updates with less than 5 microseconds of jitter.

RPU-REQ-004**Interface**

The RPU SHALL synchronize its internal control loop with the EtherCAT Distributed Clock (DC) with a maximum synchronization error of ± 1 microsecond.

Rationale: Precise synchronization with the Optimus 3 forearm and other system components is critical for coordinated multi-limb manipulation.

Verification: Test

Acceptance: EtherCAT bus analyzer confirms DC synchronization offset remains within $\pm 1\mu\text{s}$ over a 24-hour operational period.

RPU-REQ-005**Performance**

The RPU SHALL maintain a Cartesian tip position accuracy of $\pm 0.5\text{mm}$ for each finger through the inverse kinematics solver, excluding mechanical backlash.

Rationale: High-dexterity manipulation requires high precision in the mathematical mapping of task space to joint space.

Verification: Analysis

Acceptance: Numerical analysis of the IK algorithm's floating-point precision and convergence confirms a theoretical error $\leq 0.5\text{mm}$.

The RPU SHALL implement a hardware watchdog timer that initiates a safe state reset if the RTOS scheduler fails to kick the watchdog within a 2ms window.

Rationale: Ensures system reliability and safety in the event of a software hang or RTOS deadlock, supporting the 5,000-hour MTBF goal.

Verification: Test

Acceptance: Forced software hang results in a hardware reset and motor torque disablement within 2ms.

EtherCAT Slave Controller

The EtherCAT Slave Controller SHALL support a Process Data Object (PDO) cycle time of 1.0 ms with a maximum jitter of less than 1.0 μ s using Distributed Clocks (DC).

Rationale: To meet the Optimus 3 central computer's requirement for high-speed synchronous data exchange and ensure stable control of 22 degrees of freedom.

Verification: Test

Acceptance: Oscilloscope measurement of the SYNC0 pulse shows a stable 1ms period with jitter not exceeding 1.0 μ s over a 24-hour test period.

The EtherCAT Slave Controller SHALL provide a Process Data Interface (PDI) supporting SPI with a minimum clock frequency of 20 MHz for communication with the localized SoC.

Rationale: High-bandwidth internal communication is required to transfer tactile sensor data and motor commands within the 1ms control window without bottlenecking.

Verification: Analysis

Acceptance: Timing analysis confirms that the full 512-byte PDO buffer can be transferred via SPI within 250 μ s (25% of the cycle time).

ESC-003

Safety

The EtherCAT Slave Controller SHALL implement a hardware watchdog timer that triggers a 'Safe State' signal to the motor controllers if a valid EtherCAT frame is not received within 2.0 ms.

Rationale: Ensures compliance with ISO 10218-1 safety standards by preventing unpredictable behavior during communication loss.

Verification: Test

Acceptance: Disconnecting the EtherCAT cable results in the 'Safe State' signal being asserted within 2.1 ms, verified via logic analyzer.

ESC-004

Functional

The EtherCAT Slave Controller SHALL support a minimum of 4 Fieldbus Memory Management Units (FMMU) and 4 SyncManagers.

Rationale: Required to manage the complex mapping of cyclic process data and acyclic mailbox data (CoE) for 22 DoF and tactile sensors.

Verification: Inspection

Acceptance: Component datasheet confirms support for at least 4 FMMUs and 4 SyncManagers.

ESC-005

Environmental

The EtherCAT Slave Controller SHALL operate within the vibration profiles specified in MIL-STD-810H, Method 514.8, Category 4 (Truck/Trailer/Tracked).

Rationale: The humanoid platform is mobile and will experience significant mechanical stress during locomotion and manipulation.

Verification: Test

Acceptance: The component maintains error-free EtherCAT communication (zero dropped frames) during and after exposure to the specified vibration profile.

The EtherCAT Slave Controller SHOULD support Fail Safe over EtherCAT (FSoE) protocol for safety-rated torque limiting data transmission.

Rationale: FSoE provides a standardized method for transmitting safety-critical data, simplifying the path to ISO 10218-1 certification.

Verification: Demonstration

Acceptance: Successful exchange of FSoE master-slave heartbeat and safety frames between the ESC and a safety PLC.